

Project Backlog

Team Members: Meriem Belhathat, SARAH Madi, MARIA Madi

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Epic: Back-End Development

User Story 10.1: API Development for User Management

- Task 10.1.1: Set up API endpoints for user registration, login, and profile management (SARAH)
- Task 10.1.2: Implement authentication logic and token management (Meriem)
- Task 10.1.3: Develop password reset and user verification features (MARIA)

User Story 10.2: API Development for Courses and Grades

- Task 10.2.1: Create API endpoints for course management (MARIA)
- Task 10.2.2: Develop API for grade entry and retrieval (Meriem)
- Task 10.2.3: Implement API for grade calculation and finalization (SARAH)

User Story 10.3: Chatbot Integration API

- Task 10.3.1: Develop backend for chatbot NLP processing (SARAH)
- Task 10.3.2: Set up routes for chatbot communication with front-end (MARIA)
- Task 10.3.3: Test and optimize chatbot responses (Meriem)

User Story 10.4: Notification System Backend

- Task 10.4.1: Develop backend logic for notification scheduling (MARIA)
- Task 10.4.2: Create API endpoints for push notifications (SARAH)
- Task 10.4.3: Implement email notification service (Meriem)

User Story 10.5: Digital ID Card API

- Task 10.5.1: Develop backend support for digital ID verification (Meriem)
- Task 10.5.2: Set up secure storage for ID card data (MARIA)
- Task 10.5.3: Implement encryption for ID-related transactions (SARAH)

Epic: User Management and Authentication

User Story 1.1: Registration and Role Management

- Task 1.1.1: Implement registration form for all users (Meriem)
- Task 1.1.2: Develop role-based access control (RBAC) (SARAH)
- Task 1.1.3: Integrate digital ID card functionality (MARIA)
- Task 1.1.4: Implement email verification and password reset (Meriem)

User Story 1.2: Secure Login for Students

- Task 1.2.1: Integrate digital ID card reader (SARAH)
- Task 1.2.2: Implement encryption for ID card data (MARIA)
- Task 1.2.3: Ensure multi-factor authentication (Meriem)

Epic: Digital ID Card Integration

User Story 5.1: Digital ID Card Authentication

- Task 5.1.1: Implement ID card data storage and encryption (SARAH)
- Task 5.1.2: Develop ID card reader integration (MARIA)
- Task 5.1.3: Test functionality (Meriem)

User Story 5.2: Digital ID Card for Physical Resources

- Task 5.2.1: Develop QR code generation (Meriem)
- Task 5.2.2: Integrate ID verification with campus services (SARAH)
- Task 5.2.3: Test integration with external services (MARIA)

Epic: Course and Grade Management

User Story 2.1: Course Creation and Management

- Task 2.1.1: Develop course creation form (SARAH)
- Task 2.1.2: Implement update and deletion features (MARIA)
- Task 2.1.3: Develop course enrollment feature (Meriem)

User Story 2.2: Grade Entry for Teachers

- Task 2.2.1: Develop grade entry interface (MARIA)
- Task 2.2.2: Integrate grade calculation engine (SARAH)
- Task 2.2.3: Validate grade entries (Meriem)

User Story 2.3: Viewing Courses and Grades for Students

- Task 2.3.1: Develop student dashboard (Meriem)
- Task 2.3.2: Integrate grade display functionality (MARIA)
- Task 2.3.3: Implement notification system (SARAH)

Epic: Front-End Development

User Story 9.1: Student Dashboard UI/UX

- Task 9.1.1: Design the student dashboard layout (Meriem)
- Task 9.1.2: Implement dashboard using a front-end framework (SARAH)
- Task 9.1.3: Integrate API calls to display student grades and courses (MARIA)

User Story 9.2: Teacher Portal UI/UX

- Task 9.2.1: Design the teacher portal layout (SARAH)
- Task 9.2.2: Implement portal interface for grade entry (MARIA)
- Task 9.2.3: Connect API for grade submission and updates (Meriem)

User Story 9.3: Authentication Pages

- Task 9.3.1: Design login, registration, and password reset pages (MARIA)
- Task 9.3.2: Develop and style authentication pages (Meriem)
- Task 9.3.3: Integrate authentication APIs (SARAH)

User Story 9.4: Chatbot Interface

- Task 9.4.1: Design the chatbot user interface (Meriem)
- Task 9.4.2: Develop chatbot input and response components (MARIA)
- Task 9.4.3: Integrate front-end chatbot with AI backend API (SARAH)

Epic: Chatbot Assistance

User Story 3.1: Chatbot for Student Queries

- Task 3.1.1: Develop chatbot interface (MARIA)
- Task 3.1.2: Integrate AI for NLP (SARAH)
- Task 3.1.3: Train chatbot with FAQs (Meriem)

User Story 3.2: Chatbot Assistance for Teachers

- Task 3.2.1: Program chatbot for grade input guidance (SARAH)
- Task 3.2.2: Implement reminder and update features (MARIA)

Epic: Grade Appeal and Deliberation Management

User Story 4.1: Grade Appeal Submission

- Task 4.1.1: Develop appeal submission form (Meriem)
- Task 4.1.2: Integrate with course and grade DB (SARAH)
- Task 4.1.3: Create appeal tracking feature (MARIA)

User Story 4.2: Review and Deliberation by Teachers

- Task 4.2.1: Develop interface for appeal management (MARIA)
- Task 4.2.2: Implement deliberation panel (Meriem)
- Task 4.2.3: Log deliberation outcomes (SARAH)

Epic: Reporting and Analytics

User Story 7.1: Student Performance Reporting

- Task 7.1.1: Develop reporting module (MARIA)
- Task 7.1.2: Integrate charts and visualizations (SARAH)
- Task 7.1.3: Enable report export (Meriem)

User Story 7.2: Downloadable Grade Reports for Students

- Task 7.2.1: Create report download feature (Meriem)
- Task 7.2.2: Secure report storage (MARIA)
- Task 7.2.3: Test report generation (SARAH)

Epic: System Security and Compliance

User Story 8.1: GDPR Compliance

- Task 8.1.1: Implement data access and deletion features (Meriem)
- Task 8.1.2: Encrypt sensitive student information (SARAH)
- Task 8.1.3: Conduct security audits (MARIA)

User Story 8.2: Performance and Security Testing

- Task 8.2.1: Conduct load testing on APIs (SARAH)
- Task 8.2.2: Implement automated security checks (MARIA)
- Task 8.2.3: Optimize front-end performance (Meriem)